Functional Environmental Flows from Reinforcement Learning: a Proof of Concept on a Lake McClure–like Reservoir with OpenHydroQual as the Simulator

OpenHydroQual-RL working notes

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Abstract

Environmental flow (e-flow) requirements below dams are usually *prescriptive*: a fixed release schedule, chosen in advance for each water-year type. This note tests the alternative proposed in the "DRL for e-flows" outline: put the ecological objectives into the *reward* of a reinforcement-learning (RL) agent and let the agent decide, day by day, how to release water. We build a reservoir patterned on Lake McClure on the Merced River (California), drive it with the observed daily unimpaired inflow for water years 1989–2025, and simulate it with OpenHydroQual (OHQ) through its Python bindings. Each day the agent chooses the river release and the irrigation delivery. The ecological objective follows the five functional-flow components of the California Environmental Flows Framework (fall pulse, wet-season baseflow, peak flows, spring pulse and recession, dry-season baseflow). The targets are derived from the river's own unimpaired hydrology, separately for dry, moderate and wet years. We compare four operating modes on the 2017–2025 test period, which spans a record wet year, a drought and another wet year: (1) a regulatory minimum instream flow, (2) a prescribed functional-flow hydrograph, (3) functional flows in the reward, and (4) functional flows plus hydropower in the reward. Results are reported as irrigation reliability, flood days, energy and a functional-flow score between 0 and 1.

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1 Motivation and what this note does

Most e-flow requirements are *static*. A schedule of minimum flows, sometimes with pulses, is fixed in a licence and indexed to the water-year type. Static rules are easy to write down and to enforce, but they ignore information that operators actually have: the state of the reservoir, the snowpack forecast, and whether a natural storm is passing anyway. Under climate change the hydrology that a static schedule was designed for also drifts away from the hydrology it has to work with.

An *adaptive* rule maps what the operator knows today to a release today. Reinforcement learning is a general way to find such a rule by trial and error in a simulator. We specify *what* we want (the reward) and the algorithm searches for *how* to operate. The question the outline poses is whether e-flows can be expressed as part of that reward, so that an e-flow-friendly operation emerges from the optimization instead of being prescribed. The outline's Scenario 3, "reward-function functional flows", is its core novelty.

This note is a working proof of concept of that idea. It follows the outline's four scenarios on a single reservoir:

1. **Minimum instream flow** (MIF). A regulatory floor on the river flow; the agent (or rule) manages supply and floods above it.
2. **Prescribed functional flows.** A fixed functional-flow hydrograph for each year type replaces the MIF floor.
3. **Functional flows in the reward.** No floor. Every functional-flow component is scored daily and added to the reward.
4. **Multi-objective.** Scenario 3 plus hydropower in the reward (flood control is in every scenario).

The general RL background (states, actions, rewards, value functions, the PPO algorithm) is explained step by step in the companion tutorial report *Learning Reservoir Operating Rules with Reinforcement Learning* (`report/main.pdf`). Here we only summarize what is needed (Section 4) and concentrate on the e-flow problem.

2 The system

2.1 A Lake McClure–like reservoir

Lake McClure (New Exchequer Dam) on the Merced River stores about 1.02 million acre-feet (1264 hm^3). It is operated by the Merced Irrigation District for irrigation, hydropower and flood control, under FERC and other in-stream flow requirements. We do *not* attempt a faithful model of the real system (no McSwain/Merced Falls re-regulation, no evaporation, simplified head and flood rules). The aim is a stylized system with realistic hydrology, demand and scale, so that the trade-offs are real ones. Figure 1 shows the model.

2.2 Data

All data are public daily series (Table 1), fetched by `prepare_merced.py`.
Water years (WY) run October 1 to September 30. Leap days are dropped, so every year has 365 daily steps, with day $d = 0$ on October 1. The mean unimpaired runoff of the training years (WY1989–2016) is 1093 hm^3/yr and the irrigation demand is 644 hm^3/yr, so in an average year a little over half of the inflow is wanted for irrigation, and in a dry year all of it and more (Figure 2).

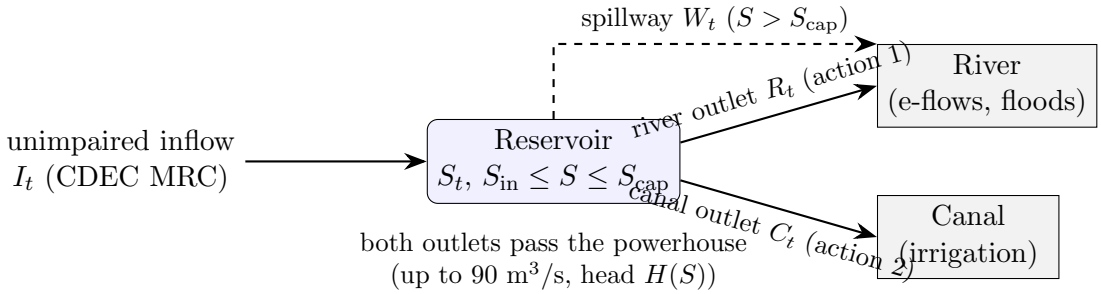


Figure 1: The model. Every day the agent sets the river release R_t and the canal delivery C_t. The spillway is uncontrolled. In OHQ the reservoir is a `Reservoir` block, the three receivers are `fixed_head` blocks whose storage accumulates the delivered volume, and the outlets are `Reservoir_link_rule` links.

Series	Source	Use	Peri
Full natural (unimpaired) flow, Merced R. nr Merced Falls	CDEC MRC, sensor 8	inflow I_t, targets	WY
Lake McClure storage	CDEC EXC, sensor 15	initial storage of test windows	1988
Lake McClure outflow	CDEC EXC, sensor 23	irrigation demand	1994
Merced R. near Snelling (below the diversions)	USGS 11272500	demand; historical river flow	WY

Table 1: Data. The irrigation demand is the observed diversion, i.e. reservoir outflow minus the river flow below the Crocker–Huffman diversion dam (WY1994–2010).

2.3 Parameters, demand, flood control and minimum flow

The **minimum instream flow** of Scenario 1 is stylized after the FERC licence and the Davis–Grunsky requirements on the Merced: 75 cfs in late October, 180 cfs from November to March, 75 cfs in April and May, and 25 cfs from June to mid-October, with 60% of these in dry years. Figure 4 shows it in blue.

3 Functional flows as numbers

3.1 The five components

The California Environmental Flows Framework (CEFF; Yarnell et al. 2020) describes a healthy hydrograph of a California river by five *functional-flow components*, each tied to ecological functions:

Fall pulse (FA). The first rains of autumn. It flushes fine sediment and cues salmon migration and spawning.

Wet-season baseflow (WB). Sustained winter flow. It keeps habitat wetted for rearing.

Peak flows (PK). Large, infrequent floods. They move the bed, scour vegetation and connect the floodplain.

Spring pulse and recession (SP). The snowmelt pulse and its gradual decline. Spawning and seed dispersal follow it, and fish are stranded if it stops abruptly.

Dry-season baseflow (DS). Low summer flow. It sets the summer habitat and water temperature.

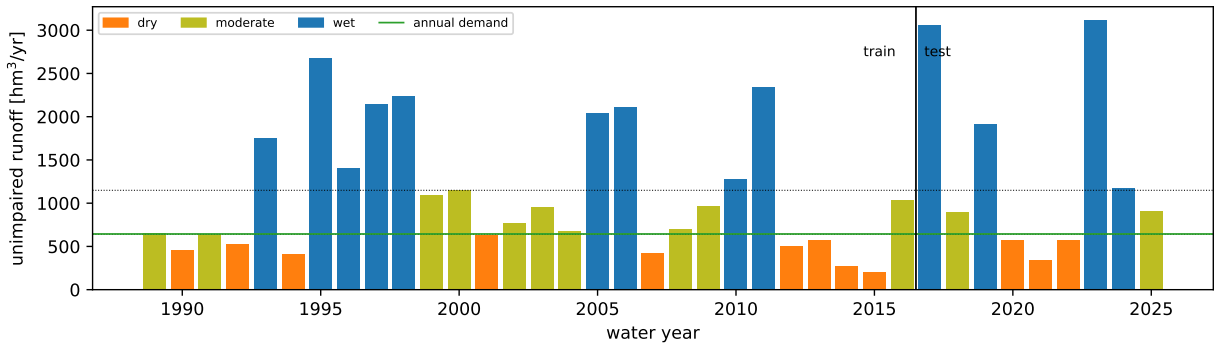


Figure 2: Annual unimpaired runoff and year types. The dotted lines are the terciles of the training years (640 and 1149 hm^3), which define dry, moderate and wet years. The green line is the annual irrigation demand. The test period WY2017–2025 contains the wettest year of the record (2023), the 2020–2022 drought and the wet 2017.

Symbol	Value	Meaning
S_{cap}	1264 hm^3	capacity (1.02 MAF); spillway opens above it
S_{in}	80 hm^3	inactive pool; both outlets taper to zero just above it (the observed minimum was 78 hm^3 in 2015)
$R_{\max}$	300 m^3/s	river outlet capacity
Q_{safe}	170 m^3/s	safe channel capacity (6000 cfs)
D_d	Fig. 3	irrigation demand on water-year day d; peak $D_{\max} = 50$ m^3/s
$S_{\mathrm{toc}}(d)$	Fig. 3	flood-control rule curve: 432 hm^3 (350 kAF) of flood space Nov 1–Mar 1
$H(S)$	70–140 m	gross head, $70+70\,x^{0.6}$ with $x = (S-S_{\mathrm{in}})/(S_{\mathrm{cap}}-S_{\mathrm{in}})$
P	$\leq$ 94.5 MW	$P = \rho g\,\eta\,\min(R+C,\,90)\,H$, $\eta = 0.88$

Table 2: System parameters.

For an RL agent, "a healthy hydrograph" must become a number it can earn every day. We give each component a calendar window, a target, and a score. CEFF detects the timing of each component from the hydrograph; we use fixed windows to keep the rules readable (Table 3).

Two choices make the score robust to small release errors. Graded credit means a release 2% short of a target loses a little, not a whole day. Measuring the recession on the trailing 7-day mean flow, $\rho_d = (\bar{Q}_{d-1} - \bar{Q}_d)/\bar{Q}_{d-1}$, matches how $r_{\max}$ is derived from the natural flow (below). A first version with pass/fail days and raw daily declines turned out to be so sensitive that day-to-day release jitter of a few percent failed most of the spring (Section 6).

The **functional-flow (FF) score** of a year is the mean of the component scores that apply. A pulse with a required duration of zero (peak flows in dry years) does not apply. A score of 1 means every component is fully met.

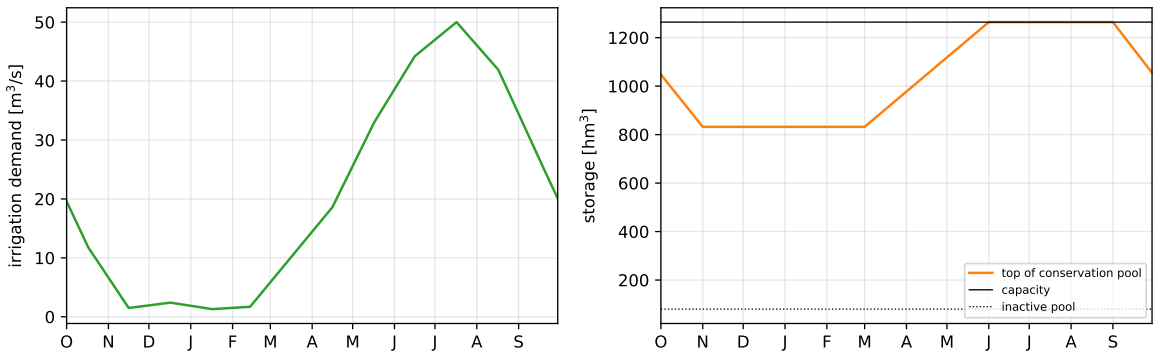


Figure 3: Left: daily irrigation demand (mean observed diversion per month, interpolated). Right: flood-control rule curve used by the rule-based baselines and given to the agent as an observation.

Component	Window	Requirement	Year score $\in [0,1]$
FA	Oct 15 – Dec 15	$\geq D_{FA}$ days with $Q \geq Q_{FA}$	$\min(1,\ \sum_d c_d / D_{FA})$
WB	Dec 16 – Mar 31	$Q \geq Q_{WB}$ every day	mean of c_d
PK	Dec 1 – Apr 30	$\geq D_{PK}$ days with $Q \geq Q_{PK}$	$\min(1,\ \sum_d c_d / D_{PK})$
SP	Apr 1 – May 31	$\geq D_{SP}$ days with $Q \geq Q_{SP}$	$\frac{1}{2}\min(1, \sum_d c_d / D_{SP})$
	Apr 15 – Jun 30	decline of the 7-day mean flow $\rho_d \leq r_{\max}$	$+\frac{1}{2}$ mean recession credit
DS	Jul 1 – Oct 14	$Q \geq Q_{DS}$ every day	mean of c_d

Table 3: Functional-flow components as used here (Q is the river flow below the reservoir, including spill). Day credit $c_d = \mathrm{clip}\big((Q_d/Q_k - 0.9)/0.09,\ 0,\ 1\big)$: full at 99% of the target or more, falling linearly to zero at 90%. Recession credit is 1 for $\rho_d \leq r_{\max}$ and falls linearly to 0 at $\rho_d = 2r_{\max}$.

3.2 Targets from the river's own natural flow

Following CEFF, targets come from the unimpaired hydrology of the same river over the training years, for each year type:

- Q_{WB}: median over the years of the window's 10th-percentile flow (CEFF `Wet_BFL_Mag_10`);
- Q_{DS}: 25th percentile over the years of the window's median flow (the lower end of the natural range of `DS_Mag_50`);
- Q_{FA}: 25th percentile of the natural fall maxima, $D_{FA} = 2$ days;
- Q_{SP}: half the median natural spring maximum; D_{SP} the median number of natural days above it, capped at 7;
- Q_{PK}: the median annual peak (about the 2-year flood) capped at $0.8\,Q_{\text{safe}}$, because a regulated river cannot be asked to flood its own channel; D_{PK} is the median number of natural days above it (capped at 7);
- $r_{\max}$: 90th percentile of the natural daily declines of the 7-day mean flow during the recession window (about 10%/day, close to the CEFF guidance for Sierra snowmelt rivers).

Table 4 gives the values and Figure 4 shows them against the natural hydrographs. The unimpaired flow itself scores only 0.76–0.83, because natural years scatter around the targets: a natural dry

year often has no fall pulse. A regulated river that scores near 0.8 is therefore about as "functional" as the natural one *by these metrics*.

Target	Unit	dry	moderate	wet
Fall pulse magnitude Q_{FA}	m^3/s	10.96	10.24	31.38
Fall pulse duration D_{FA}	d	2	2	2
Wet-season baseflow Q_{WB}	m^3/s	2.80	7.54	10.77
Peak flow magnitude Q_{PK}	m^3/s	136.00	136.00	136.00
Peak flow duration D_{PK}	d	0	1	7
Spring pulse magnitude Q_{SP}	m^3/s	44.64	75.23	120.25
Spring pulse duration D_{SP}	d	7	7	7
Max. recession rate $r_{\max}$	1/d	0.10	0.10	0.10
Dry-season baseflow Q_{DS}	m^3/s	0.34	1.67	4.11
Prescribed hydrograph volume	hm^3/yr	109	240	454
Minimum instream flow volume	hm^3/yr	53	89	89
FF score of the unimpaired flow	–	0.76	0.78	0.83

Table 4: Functional-flow targets by year type (from WY1989–2016 unimpaired flow), volume of the prescribed hydrograph and of the minimum instream flow, and the score that the unimpaired flow itself gets.

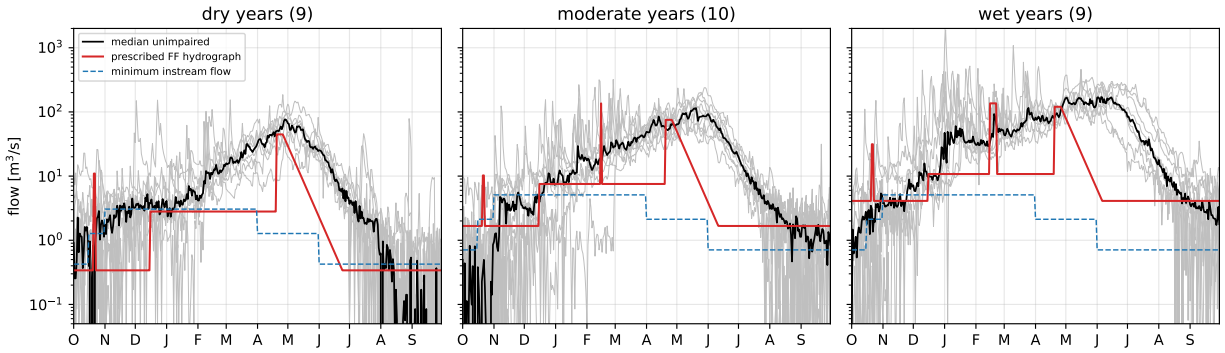


Figure 4: Unimpaired daily flow of the training years (gray; median in black), the prescribed functional-flow hydrograph used in Scenario 2 (red), and the minimum instream flow of Scenario 1 (blue, dashed). Log scale.

3.3 The prescribed hydrograph (Scenario 2)

The prescriptive alternative turns the targets into a fixed release schedule for each year type, met by construction: baseflow floors in their windows, a fall pulse on October 21, a peak-flow block on February 15 (moderate and wet years), and a spring pulse on April 20 followed by a recession at $0.8\,r_{\max}$ down to the dry-season floor (red lines in Figure 4). It costs 109/240/454 hm^3 of river water in dry/moderate/wet years, against 53–89 hm^3 for the minimum instream flow.

3.4 Year type under uncertainty

Targets depend on the year type, but the type of the current year is not known until it is over. As in California practice, the operator works with a *declared* type based on the runoff forecast. We imitate a Bulletin-120 forecast: the remaining runoff of the water year is known with a log-normal error. The error's spread falls from 50% before February to 15% in April and 5% in June, and the error is redrawn every month. On each day the declared type is the tercile class of (runoff so far + forecast of the rest). Daily targets follow that day's declaration, and the year's score uses the May 1 declaration. A prescribed schedule therefore sometimes releases a "dry" winter baseflow in what turns out to be a moderate year, which is the real cost of deciding early.

4 The reinforcement-learning problem

4.1 In one paragraph

An RL *agent* (a small neural network, the *policy*) looks at the *state* of the system each day and chooses an *action*. The simulator advances one day and returns a *reward*. Training adjusts the network so that the expected sum of discounted rewards, $\mathbb{E}\big[\sum_t \gamma^t r_t\big]$, is as large as possible. Nothing about how to operate is programmed in. Everything the agent learns comes from the reward and the simulated consequences of its own choices. The companion report explains this in detail.

4.2 What the agent sees (state)

The observation o_t has 18 entries, all scaled to order one: storage S_t/S_{cap}; today's inflow and its 7- and 30-day means (as $\log(1+I/10)$); the season (sin, cos of $2\pi d/365$); the forecast of the remaining water-year runoff/S_{cap}; yesterday's river flow; today's demand/$D_{\max}$; today's *reference* river flow (in action units: the hard floor in Scenarios 1–2, the prescribed functional-flow schedule of the declared year type in Scenarios 3–4, where it is only a hint the agent may ignore); the declared year type (three 0/1 flags); the progress of the fall, peak and spring pulses this year (days achieved/days required); the flood space $(S_{\text{toc}} - S_t)/S_{\text{cap}}$; and a spill-risk indicator $(S_t + \text{forecast} - S_{\text{cap}})/S_{\text{cap}}$. The pulse-progress entries are needed because the pulse rewards depend on what has already been achieved this year. Without them the reward would not be a function of the observed state.

4.3 What the agent decides (actions)

Two numbers $a_1, a_2 \in [-1, 1]$:

$$R_t = R_{\min}\left(\frac{R_{\max}}{R_{\min}}\right)^{(a_1+1)/2}, \qquad C_t = D_d\,\min\big(1,\ 0.6\,(a_2+1)\big), \tag{1}$$

with $R_{\min} = 0.1$ and $R_{\max} = 300$ m^3/s. The river release is on a log scale because the targets range from 0.3 m^3/s (dry-season baseflow) to 136 m^3/s (peak flow): an error in a_1 is then the same *relative* flow error at every level. An earlier version with $R_t \propto (a_1+1)^2$ could not hold small baseflows accurately enough. Delivery is a fraction of today's demand, reaching 100% at $a_2 \geq 2/3$. In Scenarios 1 and 2 the river release is raised to the floor if needed, $R_t \leftarrow \max(R_t, F_t)$. Both outlets taper to zero just above the inactive pool, so the reservoir cannot be overdrawn.

4.4 What the agent is rewarded for

With $Q_t = R_t + W_t$ the river flow (release plus spill), the daily reward is the sum of these terms:

$$r_t^{\text{sup}} = -w_s \left(\frac{(D_d - C_t)^+}{D_{\max}} \right)^2 \qquad \text{irrigation shortage (squared: many small cuts beat one big} \tag{2}$$

$$r_t^{\text{fl}} = -w_f \, \frac{(Q_t - Q_{\text{safe}})^+}{Q_{\text{safe}}} \qquad \text{flooding below the dam} \tag{3}$$

$$\begin{aligned} r_t^{\text{ff}} = & -w_{ff} \sum_{k \in \{WB, DS\}} \mathbb{1}_k(d) \, \frac{1 - c_t^{(k)}}{N_k} && \text{baseflow days missed} \\ & + \sum_{k \in \{FA, PK, SP\}} \omega_k \, w_{ff} \, \mathbb{1}_k(d) \, \Delta\min\Big(1, \frac{n_k}{D_k}\Big) && \text{pulse progress, } n_k \leftarrow n_k + c_t^{(k)} \\ & - \tfrac{1}{2} w_{ff} \, \mathbb{1}_{REC}(d) \, \frac{1}{N_{REC}} \min\left(1, \frac{(\rho_t - r_{\max})^+}{r_{\max}}\right) && \text{too-fast recession of the 7-day mean} \end{aligned} \tag{4}$$

$$r_t^{\text{hp}} = w_h \, P_t / P_{\max} \qquad \text{energy} \tag{5}$$

where $\mathbb{1}_k(d)$ is one inside the window of component k, N_k is the number of days in that window, $c_t^{(k)}$ is the day credit of Table 3, $\omega_{FA} = \omega_{PK} = 1$, $\omega_{SP} = \frac{1}{2}$, and n_k accumulates the day credits of pulse k this year.

The functional-flow reward is the score, paid daily. Summed over a year, r^{ff} equals w_{ff} times the sum of the component scores, minus a constant. Every component is worth $w_{ff} = 5$ per year, and the agent is paid for exactly what it is later judged on. An earlier version used a squared baseflow shortfall, $-\big((Q_k - Q_t)^+ / Q_k\big)^2$, instead. A 15% shortfall then cost almost nothing in reward, but it zeroes the day in the score. PPO duly learned to shave baseflows to save water: its reward rose above the rule's while its functional-flow score fell from 0.85 to 0.70. We met the same trap in the stormwater-pond problem of the companion report. When the reward and the evaluation measure differ, RL optimizes the reward.

The pulse term pays the *increase* of the progress $\min(1, n_k/D_k)$ and never pays the same progress twice. A pulse therefore earns at most $\omega_k w_{ff}$ per year, even when the declared year type (and with it D_k) changes during the year. Partial pulses earn partial credit. Two loopholes in earlier versions are worth recording, because RL finds such holes quickly. Near-miss days were paid without counting toward the duration, so hovering just below the target paid more than completing the pulse. And the payment per day was w_p/D_k with the *current* D_k, so a change of year type mid-pulse paid twice. Scenarios combine the terms as follows:

Scenario	Hard river floor	Reward
1 MIF	minimum instream flow	$r^{\text{sup}} + r^{\text{fl}}$
2 Prescribed	prescribed FF hydrograph	$r^{\text{sup}} + r^{\text{fl}}$
3 FF reward	none	$r^{\text{sup}} + r^{\text{fl}} + r^{\text{ff}}$
4 Multi	none	$r^{\text{sup}} + r^{\text{fl}} + r^{\text{ff}} + r^{\text{hp}}$

The weights are $w_s = 5$, $w_f = 1$, $w_{ff} = 5$ and $w_h = 0.1$. They are value judgements, not physics. A 15% shortage spread over the irrigation season costs about -16 per year, and raising the functional-flow score from 0 to 1 is worth $5\times$(number of components), i.e. 20–25 per year. Changing the weights traces out different points on the supply–ecology trade-off (Section 9).

To compare policies that were trained or operated under different scenarios, we use one common yardstick, J_{ff}: the mean per-year sum of $r^{\text{sup}} + r^{\text{fl}} + r^{\text{ff}}$, i.e. the Scenario 3 reward, computed for every policy.

4.5 Episodes, discounting and the end of an episode

An episode is three consecutive water years, starting from a random training year (WY1989–2016) and a random storage between 30% and 85% of capacity. Three years let carry-over storage matter, as in drought sequences. The discount factor is $\gamma = 0.995$, an effective horizon of about 200 days. That is long enough to see that water stored in winter pays off in summer.

The end of an episode is a *time limit*, not the end of the world. The reservoir would keep operating. Treating the last day as terminal would teach the agent that water left in storage at the end is worthless, and it would empty the reservoir. We therefore *bootstrap* at the time limit: the critic's estimate of the value of the final state is added to the last reward ($r_T \leftarrow r_T + \gamma\, V(s_{T+1})$). This addition to the training code (`train_ppo.py`, the `boot` entries) is the standard fix for time limits.

5 The simulator: one OHQ run per day

Every decision step builds and solves a one-day OHQ model through `openhydroqual_py` (`McClureBackend` in `ohq_eflows_env.py`):

- a `Reservoir` block with the day's storage and a constant inflow time series;
- three `fixed_head` receivers (River, Canal, Spill), whose storage accumulates the delivered volume, so day-mean flows are read exactly;
- two `Reservoir_link_rule` outlets with $Q_{\max}$ set to the action and a linear taper over the 1% of capacity above S_{in};
- a third `Reservoir_link_rule` as spillway, opening from S_{cap} to $1.01\, S_{\text{cap}}$ with a capacity of 5000 m^3/s;
- solver step 0.1 d, fixed (`max_timestep_increase_factor`=1), one solver thread per model (`n_threads`=1), because 24 models run in parallel.

The daily water balance $S_{t+1} = S_t + (I_t - R_t - C_t - W_t)\,\Delta t$ closes to about 10^{-4} hm^3, i.e. 10^{-7} of the storage. One worker runs about 150 simulated days per second. With 24 parallel workers training runs at about 1400 days per second, so four million simulated days (about 11,000 years of operation) take under an hour.

6 How the agent is trained

We use proximal policy optimization (PPO), the same actor–critic algorithm as in the companion report. An *actor* network maps o_t to the mean of a Gaussian over actions, and a *critic* network estimates the value $V(o_t)$. Training repeats: 24 workers each run the current policy for 512 days; advantages are computed with GAE ($\lambda = 0.95$); both networks are updated for 5 epochs with the clipped PPO objective (learning rate 5×10^{-5}, rewards scaled by the running standard deviation of the discounted return). Getting PPO to improve on the rule at all took several attempts. The lessons are as useful as the result:

1. **Learn a correction to the rule, not the rule itself (residual policy).** Our first attempt cloned the tuned prescriptive rule into the actor by regression and then ran PPO. The clone was never exact: its action errors of about ±6% in flow cost 5.7 of the rule's 7.85 reward points per year, and PPO only won part of that back. In the final set-up the agent's action is a *correction* added to the rule's action, $a_t = \mathrm{clip}\big(a^{\mathrm{rule}}(o_t) + \delta_\theta(o_t)\big)$, and the actor starts at $\delta_\theta \equiv 0$. Training therefore starts from the rule exactly. The correction has full authority, $\delta \in [-1, 1]$, so the agent can override the rule completely wherever that pays. This is "residual RL". The rule supplies the calendar of a functional-flow hydrograph, which is hard to learn from scratch, and RL adapts it to storage, forecasts and the passing hydrology.

2. **Exploration noise must not create penalties by itself.** PPO explores by adding Gaussian noise to the action. On a log-scale release, noise of standard deviation 0.018 is ±7% daily flow jitter. With the first, pass/fail version of the functional-flow reward, that jitter alone cost 8.6 points per year (the rule's whole reward was 7.6): mostly recession violations, because the prescribed recession ran at 95% of the limit. PPO then learned to cope with its own noise, which made the noise-free policy *worse*. The graded credit, the 7-day-mean recession and a recession at 80% of the limit cut the noise cost to 0.85 points per year at standard deviation 0.01, which is the exploration level used.

3. **Keep the best snapshot, judged on the real objective.** A checkpoint is saved every 18 updates. Each one is run *deterministically* (no exploration noise) on five training-period windows (starting 1990, 1996, 2002, 2008, 2012), and the one with the highest mean reward per year is kept. The test years are never used for any choice.

7 Baselines

Each scenario has a rule-based baseline, a *standard operating policy with hedging*:

- river: today's floor (MIF or prescribed hydrograph); in Scenarios 3–4, which have no floor, the prescribed hydrograph, so the baseline is the prescriptive operation scored with the functional-flow reward;
- flood control: storage above the rule curve $S_{\mathrm{toc}}(d)$ is released over n_{dd} days, up to Q_{safe};
- canal: full demand, cut to a fraction h_f of it when storage is below $h_S\, S_{\mathrm{cap}}$.

The three parameters (h_S, h_f, n_{dd}) were tuned for each scenario by grid search ($h_S \in \{0.2, 0.3, 0.4, 0.5\}$, $h_f \in \{0.5, 0.7, 0.85\}$, $n_{dd} \in \{5, 15\}$ d) on the same five training windows, maximizing that scenario's mean reward per year. The tuned baselines are therefore optimized for the same objectives as the agents.

8 Results

Results pending.

9 Discussion and next steps

Simplifications to keep in mind

- The system is McClure-*like*: one reservoir, no re-regulating reservoirs, no evaporation, no groundwater, approximate head and flood rule. The FERC/Davis–Grunsky flows are stylized.

- Functional-flow windows are fixed calendar windows. CEFF times the components from the hydrograph, and an agent could be rewarded for timing too.
- The demand is the same every year (mean observed diversions); real demand is higher in dry years.
- Water temperature is not modeled. Cold-water pool management is a natural extension in OHQ, which already carries heat and constituents.

Next steps suggested by this proof of concept

1. **Trade-off curves.** Repeat Scenario 3 for several w_{ff} and plot irrigation reliability against FF score. This is a Pareto front of adaptive operations to set against the single point of a prescription.
2. **Comparison with the real system.** Compare with CenSierraPywr and with historical operations, using standard reservoir performance metrics (reliability, resilience, vulnerability) by water-year class.
3. **Deep uncertainty.** Train on historical years, test on climate-perturbed inflows (warming shifts snowmelt earlier). This is where adaptive rules should beat static ones most.
4. **Algorithm.** The outline plans DQN (discrete releases). PPO with continuous actions is used here because it handles the two-dimensional, wide-range release decision directly. A DQN version with a few discrete release levels is straightforward to add for comparison.
5. **Interpretability.** Distill the trained network into a readable rule (e.g. a shallow decision tree on storage, forecast and season), so that it can be written into a licence.

Reproducing

```
python prepare_merced.py                  # download CDEC/USGS data
python plot_eflows_setup.py               # setup figures and tables
python evaluate_eflows.py --tune          # tune the rule baselines
python train_ppo.py --env eflows --env-kwargs '{"scenario": "ffreward"}' \
   --total-steps 3981312 --num-envs 24 --gamma 0.995 --lr 1e-4 \
   --normalize-rewards --reward-clip 0 --save-every 54 \
   --bc-rule ohq_eflows_env:sop_rule --bc-kwargs '<tuned ffreward params>' \
   --bc-log-std -2.3 --out results/ppo_eflows_ffreward.pt
python evaluate_eflows.py --select "results/ppo_eflows_ffreward_u*.pt" --scenario ffreward
python evaluate_eflows.py --compare       # test-period tables and figures
```

References

- Yarnell, S.M., et al. (2020). A functional flows approach to selecting ecologically relevant flow metrics for environmental flow applications. *River Research and Applications* 36(2), 318–324; and the California Environmental Flows Framework (CEFF), California Environmental Flows Working Group (2021).
- *Shifting trade-offs: finding the sustainable nexus of hydropower and environmental flows in the San Joaquin River watershed* (2022), Frontiers; the functional-flow implementation (its Section 2.3) that the e-flow paper plans to replace with an RL agent.

- Schulman, J., et al. (2017). Proximal policy optimization algorithms. arXiv:1707.06347.